Cisco IOS Command Reference

Essential switch & router commands for the Small Office IT Infrastructure Lab project

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This is a quick-reference sheet, not a full course. Commands are grouped by task so you can find what you need while working in Packet Trace, GNS3, or on real Cisco gear. Anything in `this font` is typed exactly as shown (without the prompt) at the indicated mode.

1. IOS Command Modes

Cisco IOS is organized into modes. You must be in the right mode before a command works — this is the #1 source of "Invalid input" errors for beginners.

Mode	*Prompt looks like*	*How to get there*
User EXEC	`Switch>`	Default mode after login
Privileged EXEC	`Switch#`	`enable`
Global Configuration	`Switch(config)#`	`configure terminal`
Interface Config	`Switch(config-if)#`	`interface fa0/1`
VLAN Config	`Switch(config-vlan)#`	`vlan 10`
Line Config (console/VTY)	`Switch(config-line)#`	`line console 0 / line vty 0 4`

Use `exit` to back out one level, or `end` to jump straight back to Privileged EXEC from anywhere.

2. Basic Device Setup

Command	*What it does*
`hostname SW1`	Sets the device name shown in the prompt
`enable secret cisco123`	Sets an encrypted password for privileged mode
`line console 0 / password cisco / login`	Requires a password on the console port
`line vty 0 4 / password cisco / login`	Requires a password for Telnet/SSH sessions (lines 0-4)
`service password-encryption`	Encrypts plain-text passwords in the config
`no ip domain-lookup`	Stops the device trying to DNS-resolve mistyped commands (avoids long pauses)
`banner motd # Authorized access only #`	Sets a login banner (text between the # symbols)

3. VLANs & Switching

Command	*What it does*
`vlan 10`	Creates VLAN 10 and enters VLAN config mode
`name SALES`	Names the VLAN (run inside vlan config mode)
`interface fa0/1`	Selects a specific switch port to configure
`interface range fa0/1 - 4`	Selects multiple ports at once
`switchport mode access`	Sets the port as an access port (connects to a single end device)
`switchport access vlan 10`	Assigns the port to VLAN 10
`switchport mode trunk`	Sets the port as a trunk (carries multiple VLANs, e.g. to a router or another switch)
`switchport trunk allowed vlan 10,20,30,99`	Restricts a trunk to only the listed VLANs
`show vlan brief`	Lists all VLANs and which ports belong to each
`show interfaces trunk`	Shows which ports are trunking and which VLANs are allowed

4. Router & Inter-VLAN Routing

The most common small-office setup is "router on a stick": one physical router interface, split into sub-interfaces, one per VLAN.

Command	*What it does*
`interface gig0/0.10`	Creates sub-interface .10 on the physical interface gig0/0
`encapsulation dot1Q 10`	Tags this sub-interface to VLAN 10 (must match the switch trunk)
`ip address 192.168.10.1 255.255.255.0`	Sets the gateway IP for VLAN 10 on this sub-interface
`no shutdown`	Activates the interface (interfaces are off by default)
`show ip interface brief`	Quick view of every interface's IP address and up/down status
`show ip route`	Displays the routing table

5. DHCP Configuration (on Router)

Command	*What it does*
`ip dhcp excluded-address 192.168.10.1 192.168.10.10`	Reserves addresses DHCP should never hand out (e.g. gateways, servers)
`ip dhcp pool SALES`	Creates a DHCP pool named SALES
`network 192.168.10.0 255.255.255.0`	Sets the subnet this pool serves (run inside the pool)
`default-router 192.168.10.1`	Tells DHCP clients their gateway address
`dns-server 8.8.8.8`	Tells DHCP clients which DNS server to use
`show ip dhcp binding`	Lists which devices have leased which IP addresses

6. Verification & Troubleshooting

Command	*What it does*
`show running-config`	Shows the active configuration currently in memory
`show startup-config`	Shows the configuration saved for next reboot
`ping 192.168.20.1`	Tests basic reachability to an IP address
`traceroute 192.168.20.1`	Shows the hop-by-hop path to a destination
`show cdp neighbors`	Lists directly connected Cisco devices (great for mapping unknown topologies)
`show mac address-table`	Shows which MAC addresses the switch has learned on which ports
`show interfaces fa0/1`	Detailed stats and status for a single interface
`copy running-config startup-config (or: write memory)`	Saves your configuration so it survives a reboot -- do this often

Most common beginner mistake: forgetting to save. If you close Packet Tracer or a real switch loses power without `copy run start`, all your work is gone.

7. Basic Access Control Lists (ACLs)

Useful for the "firewall rule" piece of the project, e.g. blocking Sales from reaching the IT VLAN.

Command	*What it does*
`access-list 101 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255`	Denies traffic from the Sales subnet to the IT subnet (wildcard mask, not subnet mask)

Command	*What it does*
`access-list 101 permit ip any any`	Allows everything else (ACLs deny-all by default, so this line is required)
`interface gig0/0.10`	Select the interface facing the traffic you want to filter
`ip access-group 101 in`	Applies ACL 101 to inbound traffic on this interface
`show access-lists`	Shows all configured ACLs and how many packets matched each line

8. One-Line Cheat Sheet

Command	*What it does*
`enable`	Enter privileged mode
`configure terminal`	Enter global config mode
`interface`	Configure a specific interface
`switchport access vlan <#>`	Put a port in a VLAN
`ip address`	Assign an IP to an interface
`no shutdown`	Turn an interface on
`write memory`	Save the config
`show ip interface brief`	See all interfaces at a glance

Reference sheet prepared for the Small Office IT Infrastructure Lab project. Keep this handy while working in Cisco Packet Tracer.